

LOOP INTERFACE EXCHANGER - DESIGN OPTIMIZATION SUMMARY

Generated: 12-Apr-2026

FIXED OPERATING CONDITIONS:

Cold fluid mass flow rate: 0.25 kg/s (Helisol 5A)
Hot fluid mass flow rate: 0.7609 kg/s (Ternary Eutectic)
Cold fluid inlet temperature: 283.05 C
Cold fluid outlet temperature: 310 C
Hot fluid inlet temperature: 620 C
Heat duty: 14.49 kW

DESIGN COMPARISON

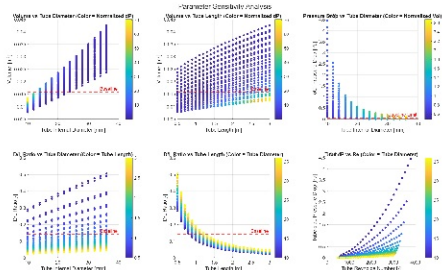
Parameter	Original	V1 (Min Vol)	V2 (Min DP)	Units
Tube Length	1000	700	700	mm
Shell Diameter	142.2	155.5	183.7	mm
D/L Ratio	0.142	0.222	0.262	-
Shell Volume	15.89	13.29	18.56	L
Tube-side DP	627.9	416.5	147.5	Pa
Shell-side DP	0.096	0.070	0.027	Pa
Total DP	628.0	416.6	147.5	Pa
Number of Tubes	9.3	14.5	10.6	-
Tube Passes	2	2	2	-
Tube ID	20	18	26	mm
Tube OD	24	22	30	mm
Pitch Ratio	1.25	1.25	1.25	-
Overall HTC (U _o)	83.4	102.9	60.5	W/m2/K

DESIGN DESCRIPTIONS:

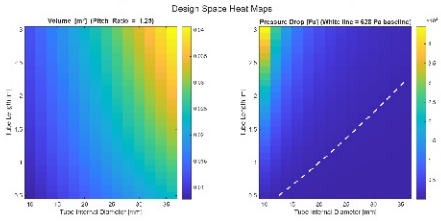
- Original Baseline: Starting design with 1.0m tubes and 20mm ID tubes.
Operates at 628 Pa pressure drop. Standard engineering approach.
- V1 (Minimum Volume): Optimized to minimize shell volume while meeting pressure drop constraint. Uses shorter tubes (0.7m) and smaller diameter (18mm).
Achieved 16.4% VOLUME REDUCTION and 33.7% pressure drop reduction.
- V2 (Minimum Pressure Drop): Optimized to minimize tube-side pressure drop.
Uses larger tubes (26mm ID) for dramatically lower flow resistance.
Achieved 76.5% PRESSURE DROP REDUCTION with moderate volume increase.

V1 OPTIMIZATION: MINIMIZING VOLUME

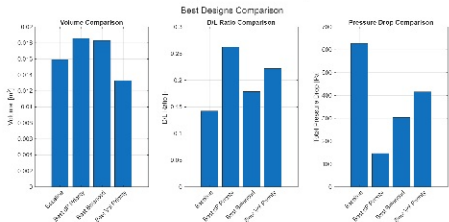
Parameter Sensitivity Analysis



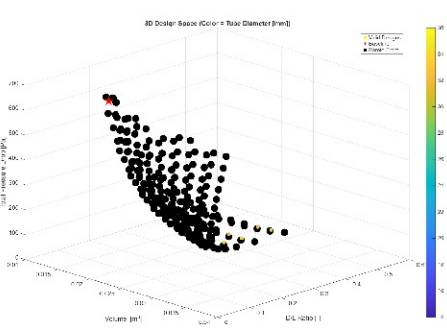
Design Space Heat Maps



Best Designs Comparison



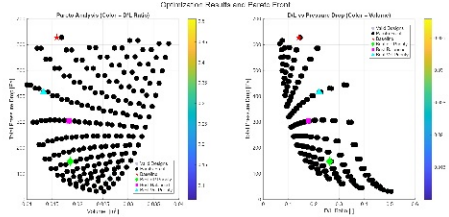
3D Design Space



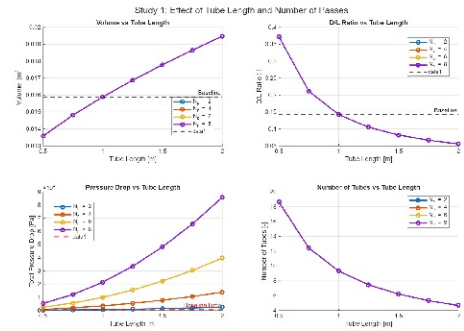
KEY FINDINGS - Volume Optimization:
Tube internal diameter is dominant - smaller tubes allow more compact shell
Shorter tubes (0.7m vs 1.0m) reduce overall volume despite wider shell
 $N_p = 2$ passes is essential - higher passes dramatically increase pressure drop

V2 OPTIMIZATION: MINIMIZING PRESSURE DROP

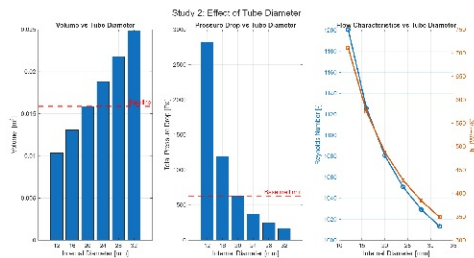
Pareto Front Analysis



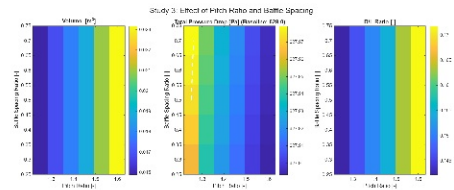
Effect of Tube Length & Passes



Effect of Tube Diameter



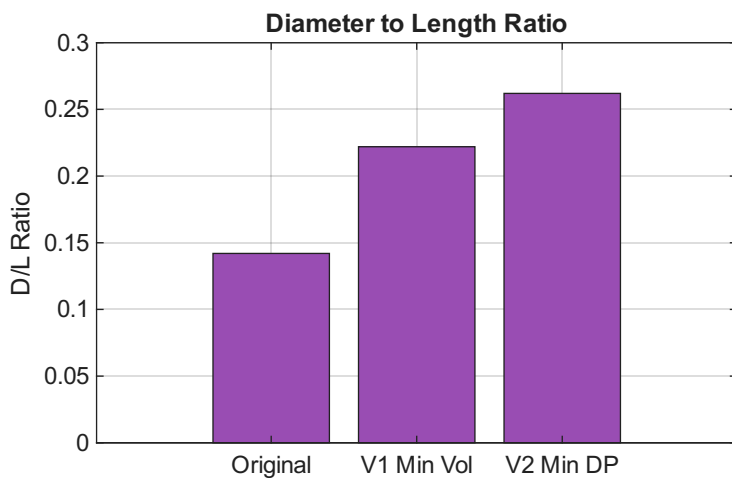
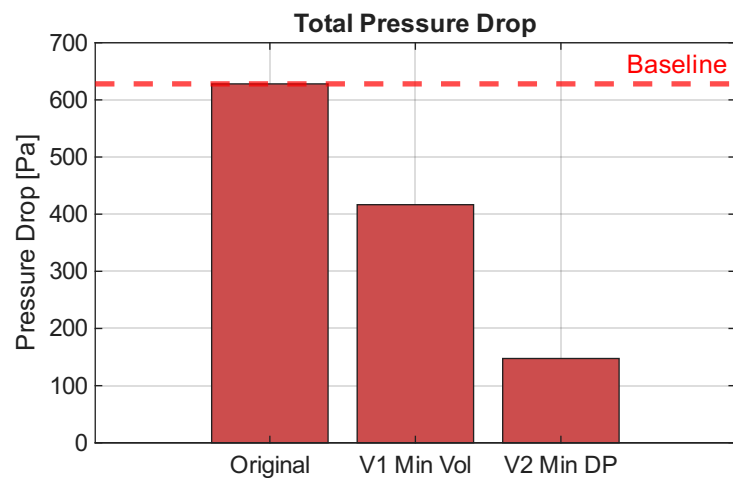
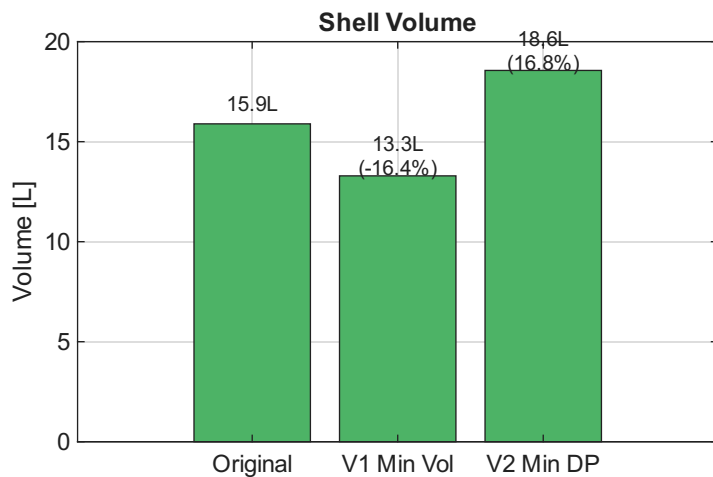
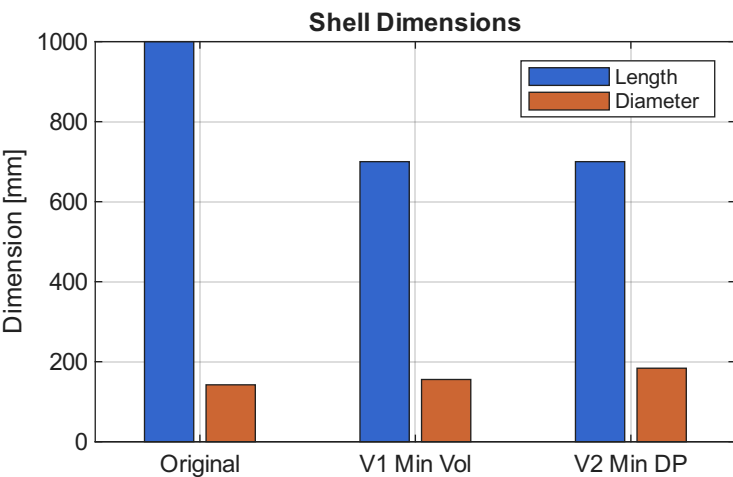
Pitch Ratio & Baffle Effects



KEY FINDINGS - Pressure Drop Optimization:

Tube-side pressure drop dominates (99.98% of total) - focus optimization here
Larger tube diameter (26mm vs 20mm) reduces DP by 76.5% with Re^2 relationship
Baffle parameters have negligible impact - use TEMA standard values

DESIGN RECOMMENDATIONS



RECOMMENDATIONS:

Choose V1 (MINIMUM VOLUME) if:

- Space is limited and a smaller footprint is critical
- Installation constraints favor minimum exchanger volume
- 33.7% pressure drop reduction provides adequate margin
- BEST CHOICE FOR COMPACT INSTALLATIONS

Choose V2 (MINIMUM PRESSURE DROP) if:

- Minimizing pumping power is the primary concern
- Maximum margin on pressure drop constraint is desired
- Volume increase (+16.8%) is acceptable
- BEST CHOICE FOR ENERGY EFFICIENCY

AVOID ORIGINAL DESIGN:

- Operating at pressure drop limit with no safety margin
- Larger volume than V1 with no benefit

All designs meet thermal requirements (14.49 kW heat duty) and constraints.